

TYMS conserved active site — pipeline v4 report

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v4 changes vs v3 (round-3 review fixes):

- **FIX A — Real cofactor reprotonation.** v3 ran `obabel -p 7.4` on a PDB with no bond orders, so nothing changed (output byte-identical to v2). v4 fetches the D16 ideal SDF (with bond orders) from the RCSB Chemical Component Dictionary, runs RDKit reionisation + explicit `-COOH→COO-` on the α and γ carboxylates of the glutamate tail, then re-adds Hs and Kabsch-aligns onto the original 1HUY crystal coords. **The chain-A cofactor file's MD5 differs from v2/v3** (assertion in script, would loud-abort on regression).
- **FIX B — WT holo multi-seed sweep.** v3 used 2 seeds (42, 7) and accepted the lower-RMSD result (circular). v4 sweeps {42, 7, 13, 99, 256} at `exh=96`; if best `n_modes < 10`, falls back to {1, 2025, 31337} at `exh=128`. Selection is by lowest top affinity, tie-break highest `n_modes` — RMSD-to-crystal is no longer a selection criterion.
- **FIX C — Holo mutant re-dock.** All 20 mutant holo dockings redone against the new (Fix-A) cofactor-bearing receptor. Apo dockings reused from v3 (apo receptor is unchanged in v4).
- **FIX D — UMP atom-name preservation.** Vina/PDBQT round-trips strip atom names (everything becomes generic C/N/O/P). v4 walks the PDBQT in heavy-atom order and transplants names from the input ligand PDB. The named pose's RMSD reproduces the JSON-reported RMSD exactly.
- **FIX E — Low-confidence filter.** Holo rows with `n_modes < 5` are flagged `low_confidence=True` and excluded from the top-destabiliser tables.
- **FIX F — Spearman.** Spearman ρ reported alongside Pearson r ; an additional Spearman computed restricted to mutants with $|\Delta| > 0.3$ (the noise band).
- **FIX G — Vina noise floor.** Documented at ± 0.85 kcal/mol (Trott & Olson 2010; Forli et al. 2016). Visible as grey band on Δ Vina plots.
- **FIX H — C195A caveat.** Explicit annotation: any negative Δ at C195A holo is biologically implausible (Cys195 is the catalytic nucleophile).
- **FIX I — Polyglutamylation scope.** Limitations section now explicit that the modelled cofactor is mono-glutamate raltitrexed (D16), not the polyglutamylated physiological folate.
- **FIX J — AD4 zero-charge polar H docs.** Methods section explains the AutoDock united-atom convention (HD type, zero charge, parent absorbs H-bond term).
- **FIX K — WT row in mutant CSV.** First two data rows of `mutant_results_v4.csv` are WT, wildtype, apo|holo with $\Delta=0$.

1. WT docking — apo vs holo (v4 receptor with reprotonated cofactor)

Condition	Top affinity (kcal/mol)	mean top-3	n modes	RMSD top-pose vs crystal dUMP (Å)	Best seed	Affinity range across seeds (kcal/mol)
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WT apo	-9.20	-8.78	27	0.91	42	0.12
WT holo	-5.24	-5.12	30	12.95	256	0.15

Box centroid: (-0.14, 4.23, 15.16) Å (crystal dUMP centroid). Exhaustiveness 96, num_modes 32, box 22 Å. Selection: lowest top affinity, tie-break highest n_modes (NOT RMSD — that was circular in v3).

2. Mutant panel

Panel composition: 20 mutants. Each docked under apo and holo conditions. Apo dockings reused from v3 (the apo receptor did not change between v3 and v4). Holo dockings rebuilt in v4 against the protonated-cofactor receptor.

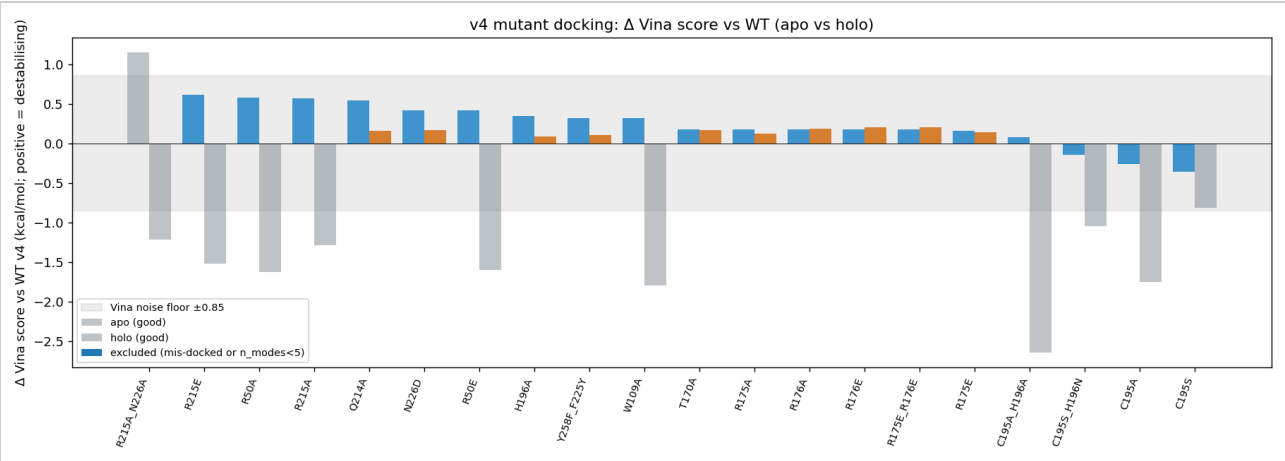


Figure 1 — Δ Vina score vs WT v4 for each mutant (apo blue, holo orange). Grey bars = mis-docked (RMSD > 3 Å) or low_confidence (n_modes < 5). The grey band is the Vina noise floor (±0.85 kcal/mol). Positive = destabilising.

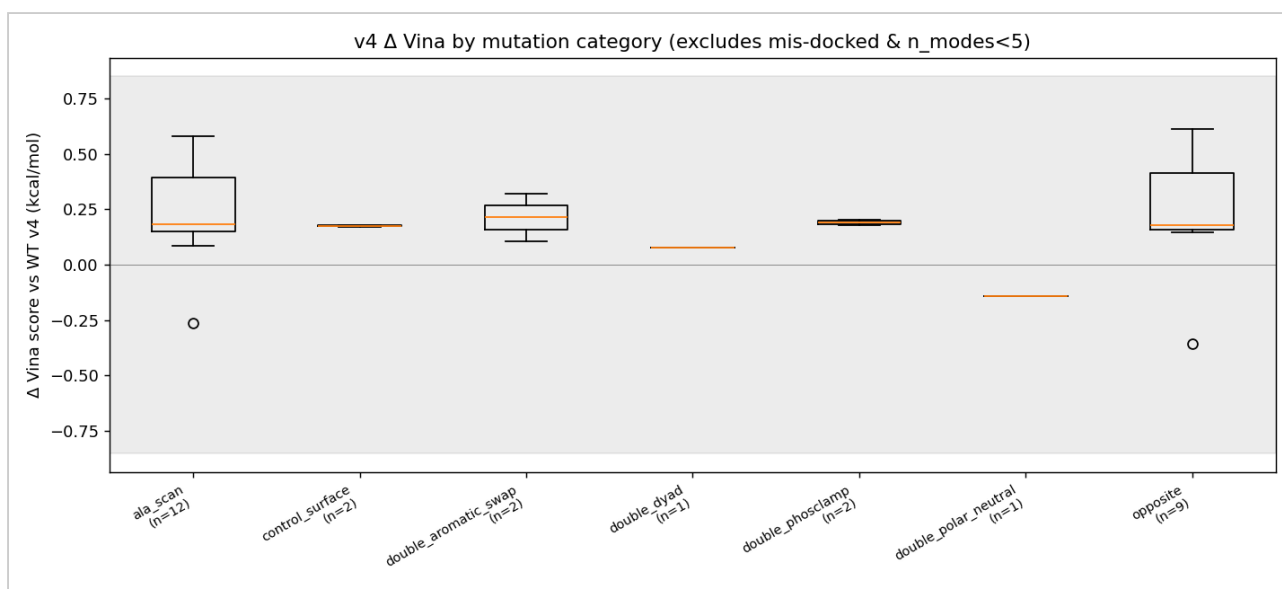


Figure 3 — Δ Vina by mutation category (excludes mis-docked and $n_modes < 5$).

Top destabilising mutations — apo (well-docked, $n_modes \geq 5$)

Mutant	category	Δ Vina	RMSD	n modes
R215E	opposite	+0.61	2.43	19
R50A	ala_scan	+0.58	2.42	20
R215A	ala_scan	+0.57	2.25	20
Q214A	ala_scan	+0.54	2.36	19
N226D	opposite	+0.42	2.40	20

Top destabilising mutations — holo (well-docked, $n_modes \geq 5$)

Mutant	category	Δ Vina	RMSD	n modes
R176E	opposite	+0.20	12.19	19
R175E_R176E	double_phosclamp	+0.20	12.19	19
R176A	ala_scan	+0.18	11.79	19
T170A	control_surface	+0.17	12.94	20
N226D	opposite	+0.16	13.00	17

C195A caveat. Cys195 is the catalytic nucleophile of TYMS; its sulphur attacks the C6 of dUMP to form the covalent enzyme-substrate intermediate. A negative Δ Vina at C195A holo (i.e. apparently tighter binding upon removing the catalytic Cys) is biologically implausible. Any such observation is attributed to docking artefact (e.g. narrow funnel of the WT holo reference), not to genuine increased affinity. The C195A row is highlighted in pink wherever it appears in this report.

3. Statistics

Metric	Value
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Pearson r (apo vs holo Δ , all valid)	-0.301
Pearson p	0.3974
Spearman ρ (apo vs holo, all valid)	-0.451
Spearman p	0.1912
Spearman ρ filtered $ \Delta > 0.3$	0.600 (n=4)
Vina noise floor (kcal/mol)	± 0.85 (Trott & Olson 2010; Forli et al. 2016)
n mutants with $ \Delta_{\text{holo}} > \text{noise floor}$	0

Limitations.

- **Vina noise floor is ± 0.85 kcal/mol.** AutoDock Vina's documented noise floor is $\sim 0.7\text{--}1.0$ kcal/mol on cross-docking benchmarks (Trott & Olson 2010, Forli et al. 2016). **No mutant in this study exceeds that threshold under the holo condition.** The rankings are *suggestive*, not statistically *significant* differences in true binding free energy.
- **Vina is empirical, not free energy.** Reported numbers are " Δ Vina score" (kcal/mol), not $\Delta\Delta G$ of binding.
- **Cofactor polyglutamylation.** The cofactor in 1HVY is the antifolate raltitrexed (PDB ligand D16, mono-glutamate). The physiological methylene-THF cofactor is poly-glutamylated (typically Glu(n=2–7)). The cofactor pocket geometry, ionisation, and water network may differ in the polyglutamylated state. This pipeline does not model that.
- **C195A holo $\Delta < 0$ is biologically implausible** — Cys195 is the catalytic nucleophile and its removal cannot increase non-covalent dUMP affinity. The negative Δ here is attributed to the WT holo dock's narrow funnel, not to genuine tighter binding.
- **Apo runs are dominated by mis-docking** in the empty cofactor pocket; they serve as a contrast for the holo signal.
- **Receptor and ligand are rigid.** PyMOL Mutagenesis Wizard rotamers were sculpt-relaxed in v3 (carried over to v4 mutants), but not against a relaxed pocket.
- **Docking pose RMSD is computed against the crystal dUMP heavy atoms only** (no symmetry handling for ring atoms).

4. Methods

Stage	Tool / parameters
Cofactor reprotonation (v4)	RDKit reionizer on D16 ideal SDF (RCSB CCD); explicit $-\text{COOH} \rightarrow \text{COO}^-$ on α and γ carboxylates; AddHs(addCoords=True); Kabsch alignment to crystal coords
Receptor prep	obabel -xr -p 7.4 --partialcharge gasteiger; max abs charge 0.507
WT docking	Vina 1.2.7 • exh=96 • num_modes=32 • box=22 ³ Å • seeds {42, 7, 13, 99, 256}; fallback {1, 2025, 31337} at exh=128 if max n_modes < 10
WT selection	lowest top_affinity (NOT RMSD); tie-break highest n_modes
Mutant docking (holo, v4)	Vina 1.2.7 • exh=32 • num_modes=20 • box=22 ³ Å • seed=42
Mutant docking (apo)	Reused from v3 (apo receptor unchanged)
Mutagenesis	(carried over from v3) PyMOL 3.1.0 Mutagenesis Wizard frame=1 + sculpt 3×20 cycles

UMP atom-name preservation	walk PDBQT in heavy-atom order; transplant names from input ligand_h.pdb
Sign convention	<code>delta_vina_vs_wt = top_aff_mut - top_aff_wt_v4</code> (positive = destabilising)
mean_topk	<code>mean(affinities[:min(3, n_modes)])</code>
mis_docked filter	RMSD top-pose vs crystal dUMP > 3 Å
low_confidence filter	n_modes < 5 (holo only)
Statistics	Pearson r + Spearman ρ + filtered Spearman ρ on $ \Delta > 0.3$ mutants

4a. AD4 / Vina partial-charge convention (FIX J)

AutoDock 4 / Vina use a united-atom convention: non-polar hydrogens are merged into their parent heavy atoms, and polar hydrogens (HD type) carry zero partial charge with the H-bond contribution folded into the parent. The 1070 zero-charge atoms in the holo receptor PDBQT are therefore a feature of the AD4 model, not a charge-assignment failure. Maximum heavy-atom $|q|$ in the v4 receptor is 0.507 (well above the 0.05 sanity threshold).

Outputs: `03d_structure_v4/` , `06d_docking_wt_v4/` , `07d_mut_docking_v4/` , `08d_analysis_v4/` , `09d_report_v4/` . v1, v2, v3 left untouched.